

Siva Subramaniyen M

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🔗 SivaSubramaniyenM

Personal Profile

M.Tech AI Engineering student with hands-on experience building end-to-end ML pipelines across deep learning, NLP, and computer vision. Backed by 2 years of industrial engineering experience demonstrating strong problem-solving, cross-functional collaboration, and the ability to deliver results independently. Eager to apply AI/ML expertise to real-world product innovation as a driven and adaptable team contributor.

Skills

Languages: C/C++, Python, SQL

ML / DL Frameworks: TensorFlow, Keras, PyTorch, Scikit-Learn, Pandas, NumPy.

Mathematics: Probability, Statistics, Linear Algebra.

GenAI: RAG (exploring), LangChain (exploring).

Cloud: AWS — EC2, S3, RDS, EBS (exploring).

Data & Visualisation: Power BI.

Dev Tools: Git, GitHub, VS Code, Jupyter Notebook, Postman, Linux (Ubuntu).

Projects

Multi-Architecture Deep Learning Pipeline — CIFAR-10 🏠 *GitHub* 2026

- Built CNN (87.04%, 4.2 ms), ResNet18 (94.55%), ViT Teacher (80.12%), and TinyViT via **Knowledge Distillation** (77.67%, 3.7 ms); also implemented DCGAN and Autoencoder; deployed via **Gradio** on Colab.
- Applied **ONNX + INT8 quantization** — compressed TinyViT from 3.2 MB to **0.24 MB** at **1.79 ms** latency (93% size reduction, zero accuracy loss). Tech: Python, PyTorch, TensorFlow, ONNX.

TextileVisionAI — Defect Detection & Fabric Classification 🏠 *GitHub* 2026

- Dual-model CV system: **YOLOv8** for real-time tear defect detection and **ResNet18** for Indian fabric pattern classification — deployed as a unified Streamlit app.
- Tech: Python, PyTorch, YOLOv8, ResNet18, OpenCV, Streamlit.

ECS-XAI — Shortcut Learning Detection in CNNs 🏠 *GitHub* 2026

- Designed novel **Explanation Consistency Score (ECS)** metric to detect shortcut learning in CNNs; tested on Colored MNIST (3 shortcut variants) using **LIME, SHAP, Grad-CAM**; documented in IEEE-format paper.
- **Led 3-member team** — built shortcut injection pipeline and formulated the metric. Tech: Python, PyTorch, LIME, SHAP, Grad-CAM.

Digital Jail — Machine Downtime Classification 🏠 *GitHub* 2025

- **Naive Bayes classifier** on a **self-collected real-world PLC alarm log dataset** (independently gathered from industrial production line) to classify machine downtime as fault vs. legitimate — with live Streamlit monitoring dashboard.
- Tech: Python, Scikit-Learn, Pandas, NumPy, Streamlit.

Alzheimer's Disease Classification — DenseNet169 🏠 *GitHub* 2025

- End-to-end medical imaging pipeline using **DenseNet169 transfer learning** to classify brain MRI scans into 4 Alzheimer's stages — achieved **~90% AUC**. Tech: Python, TensorFlow, Keras, OpenCV.

Book Recommendation System — NLP Pipeline 🏠 *GitHub* 2025

- Hybrid NLP pipeline: **VADER sentiment analysis + LDA topic modelling** to recommend books by reader sentiment and topical interest — Streamlit app with **5/5 pytest pass rate**. Tech: Python, NLTK, Scikit-Learn.

Experience

AI Prompt Engineer Intern

EDVDO

Remote

Mar 2025 – Jun 2025

- Designed and engineered prompts from scratch for an AI-powered student counselling chatbot, integrating GPT-3.5 and GPT-4 via OpenAI API to deliver contextually accurate and educationally relevant responses.
- Tested and validated prompt performance using Postman and Python-based API calls — iteratively refining prompts to improve response quality, consistency, and tone appropriate for student interactions.
- Collaborated with the product team to align chatbot outputs with real counselling workflows and student needs.

Electrical Maintenance Engineer

HL Mando Automotive India Pvt Ltd

Chennai, India

Jun 2023 – Feb 2025

- Trained in plant-wide power distribution and substation operations — gaining end-to-end understanding of industrial power infrastructure from source to end-user delivery.
- Managed machine breakdown report generation and presentation, handled spare parts inventory and procurement — developing strong data recording and analytical skills.
- Diagnosed and resolved PC, HMI, and network-related breakdowns across production lines; independently executed and monitored NAS installation across 7 assembly lines — reducing PC-related downtime by 10–15% over a 6-month rollout with zero line interruptions.
- Recognised with the plant-wide quarterly “**Spotlight**” award for exceptional independent execution of the NAS implementation project.

Education

Amrita Vishwa Vidyapeetham, Coimbatore

M.Tech in Artificial Intelligence Engineering

June 2025 – Present

- GPA: 7.3/10
- Relevant Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Explainable AI.

Thiagarajar College of Engineering, Madurai

B.E. in Electrical and Electronics Engineering

June 2019 – May 2023

- Overall GPA: 7.64/10

Certifications

- **Python Data Structures & Algorithms with LeetCode Exercises** — Udemy (2025)
- **Complete Computer Vision Bootcamp with PyTorch & TensorFlow** — Udemy (In Progress)
- **AWS Certified CloudOps Engineer Associate** — Udemy (In Progress)
- **Fullstack Web Development (MERN)** — Livewires Institute (2023)